



Downstream centrality and industry lobbying in production networks

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Abstract

Which industries lobby the most? I study whether an industry's position in the production network is associated with reported federal lobbying. The argument is that supplying many downstream customers creates two forces. A policy that helps a supplier can raise privately appropriable stakes when downstream benefits are partly capitalized back to the supplier through derived demand, supplier surplus, contracts, or preservation of customer relationships. The same downstream reach also expands the set of beneficiaries that can free-ride. The model therefore predicts a hump-shaped relationship between downstream reach and focal-industry lobbying. Using a panel of 71 U.S. industries from 1999 to 2023, I find a nonmonotone pattern in mapped LDA lobbying expenditures. The pattern appears in every annual cross-section and is stronger on spending among active industries than on lobbying entry. The estimates are descriptive and apply first to mapped spending. Only about one-third of reported spending maps cleanly to industries, and partial-identification bounds show that inference becomes weaker once the unmapped remainder is taken into account.

Keywords Lobbying · Production networks · Collective action · Partial identification · Input–output linkages

JEL Classification D72 · D85 · L14 · C14 · C23

1 Introduction

Political influence does not stop at firm or industry boundaries. Taxes, trade rules, environmental standards, and procurement decisions propagate through supplier–customer links, so the political stakes from a policy change depend in part on where an industry sits in the production network. This paper asks whether industries with broader downstream reach also report more federal lobbying expenditures.

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The answer need not be monotone. Greater downstream centrality can raise the return to lobbying when some of the benefits to customers are privately appropriable by the supplier through derived demand, supplier surplus, contracts, or preservation of customer relationships. But it can also make collective action harder because more downstream beneficiaries can free-ride on the supplier's effort. I argue that these forces generate a hump-shaped relationship: lobbying rises with downstream centrality for moderately connected industries and then falls once coordination frictions dominate.

The argument is specific to policies targeted at industry i . It does not require upstream links to be irrelevant for political organization. A policy that expands activity in i may also benefit upstream suppliers, and those suppliers could face their own collective-action problem. I therefore treat that channel as outside the model and focus on a narrower targeted-policy setting. When i supplies many downstream users, a favorable policy for i can benefit customers that need not contribute to the lobbying campaign. Upstream exposure may matter for the cost or importance of policy to i , but in this setting it does not create the same customer beneficiary set that can free-ride on i 's effort.

I evaluate this prediction descriptively in a balanced panel of 71 U.S. industries from 1999 to 2023. In the joint specification, the downstream quadratic is negative and significant, while the upstream quadratic is small and imprecise. The binned scatterplot and all 25 annual cross-sections show the same sign pattern. The hump is much stronger on the spending margin than on the entry margin.

The main data constraint is outcome coverage. Only about one-third of reported lobbying spending maps cleanly to industries, so this finding is evidence about observed mapped spending. To examine what the pattern implies for total reported lobbying, I treat the unmapped portion as an attribution problem rather than a missing-network problem, because the production network is observed for all industries in all years. That structure makes it possible to supplement the matched-data regressions with observability corrections and with structured bounds for overall eigenvector centrality, the regressor that is most balanced and most stable to coverage loss. Under economically motivated allocations, negative curvature remains, although the formal bounds interval includes zero.

The paper contributes to public choice research on the organization of political influence. In this usage, public choice refers to economic work on how incentives, collective action, rent seeking, and regulation shape political allocation. This literature is related to political science work on lobbying institutions, access, and networks among political actors, but the two literatures are not identical.¹ The classic rent-seeking and regulation literature studies how groups use the political process to obtain transfers and regulatory advantages (Becker 1983; Peltzman 1976; Stigler 1971; Tullock 1967). A related collective-action and lobby-formation tradition asks which groups can organize and which potential beneficiaries remain unorganized (Denzau and Munger 1986; Esteban and Ray 2001; Mitra 1999; Olson 1965). Empirical work on lobbying and political activity emphasizes both the determinants of group participation and the difficulty of measuring influence from observed political expenditures (Ansolabehere et al. 2003; Bombardini and Trebbi 2020; de Figueiredo and

¹ The distinction is one of emphasis rather than a disciplinary boundary. Public choice work typically treats lobbying as part of political allocation and asks how economic incentives, group formation, and collective-action constraints shape political behavior. Political science work on lobbying often asks complementary questions about institutions, access, representation, and relations among political actors. The input-output network in this paper is used to study the incentive and collective-action side of lobbying.

Richter 2014; Grier et al. 1994; Potters and Sloof 1996; Tripathi et al. 2002). Work on trade policy also shows that political organization can occur individually or through associations (Bombardini 2008; Bombardini and Trebbi 2012). This paper adds a production-network margin to the public choice question of which groups organize and which policy benefits can be privately appropriated.

The paper also relates to work on production networks and political preferences. Supplier-customer links shape trade preferences and political influence (Betz and Hummel 2025; Kim 2017; Osgood 2018), drawing on the broader view of production networks as conduits for propagated shocks (Acemoglu et al. 2012; Antras and Chor 2013; Carvalho 2014). The closest adjacent paper is Betz and Hummel (2025), which studies production networks and influence in trade politics rather than lobbying expenditure. This paper brings production-network position to the study of industry lobbying and pairs the mapped-data patterns with a partial-identification exercise for the central attribution problem created by incomplete industry assignment.²

2 Theory

The theory treats the production network as an exogenous environment and asks how a policy targeted at industry i changes the incentives of i and its downstream beneficiaries to finance a common lobbying effort. It provides a compact framework for the main empirical predictions: a hump in downstream centrality, weaker curvature in upstream centrality, a broad hump in overall centrality, and a stronger response on the intensive margin. The setup combines Olson-style collective action (Olson 1965), formalized in the group-size literature by Esteban and Ray (2001), with an exogenous production-network environment (Acemoglu et al. 2012; Antras and Chor 2013; Carvalho 2014).

2.1 Network environment and policy incidence

Let A denote a directed input–output matrix, where A_{ij} is the share of industry j 's input purchases sourced from industry i . A policy targeted at supplier i propagates through its customer chain. For $0 < \delta < 1$, define discounted downstream incidence as

$$\omega_{ji} = \sum_{k=1}^{\infty} \delta^{k-1} (A^k)_{ij}, \quad (1)$$

and let downstream reach be

²Political science also uses network methods to study relations among political actors and organizations, including general network analysis in political science and international relations (e.g. Hafner-Burton et al. 2009, Ward et al. 2011) and information or influence networks among interest groups (e.g. Carpenter et al. 2004, Heaney 2014). Those papers study networks of political actors or organizations. Here, links are input–output relationships among industries, and the question is how economic incidence affects the private appropriability of lobbying benefits.

$$D_i = \sum_{j \neq i} \omega_{ji}. \quad (2)$$

Analogously, let U_i denote an upstream exposure measure computed from A' . Downstream reach expands the set of industries that benefit when i secures favorable policy. Upstream exposure can raise the importance of the policy for industry i . It may also matter for upstream suppliers when policy for i changes demand for their output. I treat this supplier-side propagation as outside the model and focus on the targeted-policy channel in which downstream customers are the clearest passive beneficiaries of i 's lobbying.

2.2 Lobbying game and equilibrium

Consider a policy targeted at industry i . If the policy is favorable, industry i receives private gross benefit $b_i(D_i, U_i)$, where b_i is increasing in downstream and upstream exposure and weakly concave in D_i . The downstream term in b_i captures appropriable customer-side gains. It requires only that some downstream benefits return to the supplier through derived demand, supplier surplus, long-term contracts, or the preservation of customer relationships. Each downstream beneficiary j receives spillover benefit $\tau\omega_{ji}$, where $\tau > 0$ scales how much of the policy gain is transmitted through the customer chain. Let industry i and its downstream beneficiaries simultaneously choose lobbying contributions. If total lobbying for policy i is

$$G_i = g_i + \sum_{j \neq i} g_{ji},$$

policy succeeds with probability

$$p(G_i) = 1 - e^{-G_i}.$$

Contributor k with benefit v_{ki} solves

$$\max_{g_{ki} \geq 0} v_{ki}p(G_i) - \frac{1}{2}g_{ki}^2,$$

where $v_{ii} = b_i(D_i, U_i)$ for the focal industry and $v_{ji} = \tau\omega_{ji}$ for downstream beneficiary j . The first-order condition for any interior contributor is

$$g_{ki} = v_{ki}e^{-G_i}.$$

Summing across contributors gives

$$G_i = B_i e^{-G_i}, \quad B_i = b_i(D_i, U_i) + \tau D_i.$$

Hence

$$G_i = W(B_i), \quad (3)$$

where $W(\cdot)$ is the principal branch of the Lambert- W function. The focal industry's own equilibrium lobbying is therefore

$$g_i(D_i, U_i) = b_i(D_i, U_i) \frac{W(B_i)}{B_i}. \quad (4)$$

Equation (4) is the main theoretical object for the empirical analysis. Downstream reach raises total coalition stakes through B_i , but it also lowers the focal industry's share of equilibrium lobbying because $W(B_i)/B_i$ is positive and decreasing in B_i . This decomposition separates two forces. The first is a stakes effect. A more downstream-central industry affects more customer industries, and some of that customer-side value can return to the supplier. For example, a rule that preserves production of a widely used intermediate can protect the supplier's own surplus while also benefiting customers that avoid higher input costs or disruption. The second is a private-appropriation effect. As downstream reach expands, more beneficiaries can wait for others to pay for the lobbying campaign. Industry i then finances a smaller share of the equilibrium effort even if total coalition stakes continue to rise.

2.3 Proposition

Fix U_i . Suppose $D \mapsto b_i(D, U_i)$ is continuous, nonnegative, increasing near zero, and satisfies $b_i(D, U_i) = o(D/\log D)$ as $D \rightarrow \infty$. If $g'_i(0, U_i) > 0$, then focal-industry lobbying initially rises with downstream reach, satisfies $g_i(D_i, U_i) \rightarrow 0$ as $D_i \rightarrow \infty$, and therefore attains an interior maximum at some $D_i^* > 0$. The Electronic supplementary material gives a complete proof. The proposition gives a sufficient condition for a hump that comes from coalition incentives rather than from an imposed quadratic. Total coalition stakes B_i can keep rising with D_i even while the focal industry's own spending eventually declines.

The same logic implies stronger concavity in downstream centrality than in upstream exposure. Downstream reach affects lobbying through two channels: private stakes and coalition expansion. Upstream exposure affects lobbying only through the focal industry's own benefit $b_i(D_i, U_i)$ in the model. This is a maintained asymmetry, not a general theorem about policy incidence. If policy for industry i raises demand for upstream suppliers, those suppliers could have separate incentives to organize. The empirical test therefore asks whether the downstream margin, where the model's free-rider channel is most direct, has stronger curvature than the upstream margin. If the policy were instead targeted at an upstream supplier, then that supplier would become the focal industry and the same downstream logic would apply to its own customer chain.

2.4 From a policy opportunity to annual spending

The model describes a single policy opportunity, while the data measure annual LDA expenditures aggregated across many issues. I interpret annual spending as the sum of issue-level lobbying decisions faced by an industry during a year. The empirical prediction carries over if downstream reach affects either the expected stakes of those opportunities or the share of opportunities in which supplier-customer spillovers create appropriability and free-riding tradeoffs. The link to annual spending is therefore an aggregation argument. It motivates

descriptive hypotheses about annual mapped spending rather than causal claims about a particular policy intervention.

2.5 Empirical implications and observability

If becoming politically active requires paying a fixed cost F_i , industry i appears in the lobbying data only when $g_i(D_i, U_i) \geq F_i$. The extensive margin is then a thresholded version of the same underlying object, so the hump should appear more sharply in spending among active industries than in a binary lobbying indicator. The same coalition logic can also affect what is observed in the data. Suppose only a fraction $q_i(D_i)$ of the focal industry's political activity is observed as directly attributable industry spending, with $q'_i(D_i) \leq 0$ because broad downstream coalitions increasingly act through trade associations, umbrella groups, or other shared vehicles that are harder to map to industries. Observed mapped lobbying is then

$$m_i(D_i, U_i) = q_i(D_i)g_i(D_i, U_i).$$

If q_i is nonincreasing, mapped lobbying peaks weakly earlier than true focal-industry lobbying and the mapped share declines with downstream reach. This extension is useful for interpreting the data because direct industry attribution may become harder in the upper tail of downstream centrality, but it does not recover the unobserved mass.

The theory organizes five empirical hypotheses:

- H1** Mapped industry lobbying is hump-shaped in downstream centrality. Section 4 studies this prediction in the baseline downstream specification.
- H2** The same nonlinearity is weaker for upstream centrality than for downstream centrality. The joint downstream-upstream specification provides the main comparison.
- H3** The hump is stronger on the spending margin than on the binary entry margin. Table 4 compares these two margins.
- H4** A broad overall centrality measure can inherit the same general shape, but it is supporting evidence rather than the sharpest mechanism test.
- H5** Incomplete industry attribution can weaken inference about total lobbying. Section 5 therefore treats unmapped spending as an allocation problem.

3 Data

The production network comes from the annual BEA input–output use table at the summary-industry level. These tables are the source for production-network links, so they determine the level at which the empirical data can be combined. I therefore aggregate lobbying and economic variables to BEA summary industries. The empirical panel contains 71 industries from 1999 to 2023, for 1775 industry-year observations. The unit of analysis is the industry-year. I use eigenvector centrality as the benchmark overall measure because it aggregates multistep linkages, and I construct separate downstream and upstream centrality measures for the mechanism test. All centrality measures are normalized within year to mean one so that changes over time reflect relative position inside each annual network rather than level shifts in the matrix.

Lobbying expenditures come from harmonized LDA filings in LobbyView (Kim 2018). I use LDA spending because it provides a consistent federal measure of reported lobbying expenditures over the full sample period. This measure excludes state lobbying, campaign contributions and PAC activity, informal access, grassroots activity, and political work that is not reported or not directly attributable to a mapped client. I aggregate filing-year spending to the client-year level and map clients to BEA summary industries through a conservative NAICS-to-BEA concordance. Clients with ambiguous or missing NAICS are not assigned to industries. The main outcome is therefore

$$Y_{it} = \log(1 + \text{mapped spending}_{it}),$$

where mapped spending is the portion of reported lobbying that can be attributed to industry i in year t .

The aggregation from clients to industries is important for interpretation. LobbyView begins with client-level filings, but the empirical object in this paper is industry-level mapped spending. Firms within the same BEA industry can differ in their customer networks and in their own incentives to lobby. The analysis does not observe firm-specific production-network positions, so the mechanism should be read as an industry-level relationship or as an aggregation of heterogeneous firm incentives within BEA industries.

Table 1 reports the coverage rates in the lobbying data. Only 32.21% of total reported lobbying spending is mapped to industries in the baseline concordance. Coverage varies over time but is not concentrated in a few years: the mapped share is 27.43% in 1999, 43.46% in 2009, 45.07% in 2017, and 30.43% in 2023. Coverage also shows little relation to centrality. Average coverage by centrality quartile is 32.42%, 32.81%, 31.71%, and 31.57%, which makes it less likely that the headline result is driven only by the most central sectors.

Table 1 Sample structure, mapping coverage, and core variables

Panel A. Mapping coverage				
Statistic	Value			
Overall mapped-spending share	32.21%			
Yearly mapped-share minimum	25.97% (2003)			
Yearly mapped-share maximum	45.07% (2017)			
Mapped share in 1999	27.43%			
Mapped share in 2009	43.46%			
Mapped share in 2023	30.43%			
Coverage mean, centrality quartile 1	32.42%			
Coverage mean, centrality quartile 2	32.81%			
Coverage mean, centrality quartile 3	31.71%			
Coverage mean, centrality quartile 4	31.57%			
Panel B. Core variables (industry-year panel)				
Variable	Mean	SD	p25	p75
Eigenvector centrality	1.000	0.519	0.685	1.140
log (1 + mapped lobbying)	12.587	6.338	12.569	16.582
Indicator for positive mapped spending	0.810	0.393	1.000	1.000
log (Value Added)	11.749	1.153	10.989	12.546

Unit of observation in Panel B is industry-year (71 industries, 25 years, $N = 1775$). Centrality is normalized within year to mean one. The p25 and p75 columns report the 25th and 75th percentiles. Coverage quartiles use the industry-level coverage proxy implied by yearly mapped shares

These balance statistics suggest that mapped and unmapped observations differ in ways that matter for interpretation. Table S1 in the Electronic supplementary material reports standardized differences between covered and uncovered industry-years. The difference is small for eigenvector centrality (0.109), large for downstream centrality (0.835), and moderate for log value added (-0.391). That pattern matters because the eigenvector regressor looks relatively balanced across covered and uncovered observations, while downstream centrality does not. The mapped-data downstream result is therefore best read as the main observed pattern. It is not, by itself, a claim about total lobbying after the unmapped mass is assigned. Section 2 suggests one economic reason for that imbalance: industries with high downstream centrality may increasingly lobby through trade associations or other shared vehicles. Such organizations can appear as separate clients, and their expenditures can be difficult to allocate to member industries. The observed upper-tail decline in mapped spending may therefore partly reflect organizational form rather than lower total political activity.

The network covariates are better measured than the outcome. Centrality is observed for every industry-year, not only for covered observations. Recomputing eigenvector centrality on the covered subnetwork yields a correlation of 0.977 with the full-network measure, so the benchmark regressor is stable under coverage loss. The analogous downstream directional measure is less stable, with Pearson correlation 0.808 and Spearman correlation 0.893. For this reason, the formal bounds exercise uses the benchmark eigenvector regressor and treats the directional downstream exercise as a secondary coverage check.

The attribution problem has a simple structure. For each year, the data provide mapped spending by industry, total reported spending, and network position for every industry. What is unknown is how the unmapped block is distributed across industries. The bounds exercise therefore uses observed industry size, mapped spending, and network position to restrict the feasible assignments of unmapped spending. The Electronic supplementary material formalizes this allocation problem and shows how those observed restrictions discipline the later bounds exercise.

4 Results

The regressions below describe conditional associations between network position and mapped lobbying spending. Industry and year fixed effects absorb time-invariant industry differences and aggregate shocks, but they do not identify exogenous variation in centrality. Policy salience, concentration, firm composition, or earlier lobbying could be correlated with both production-network position and reported spending. The leave-one-year-out and annual cross-sectional exercises reported below are temporal stability checks, not dynamic identification strategies.

The baseline specification is

$$Y_{it} = \alpha_i + \tau_t + b_1 \text{Centrality}_{it} + b_2 \text{Centrality}_{it}^2 + \gamma \log(\text{VA}_{it}) + \varepsilon_{it}, \quad (5)$$

In this notation, Centrality_{it} denotes the network measure used in a given specification, either downstream, upstream, or overall eigenvector centrality, and VA_{it} is industry value added. All specifications include industry and year fixed effects, and standard errors are clustered by industry.

4.1 Downstream hump

Table 2 reports the main regressions. Column (1) estimates the downstream specification on its own. Column (2) does the same for upstream centrality. Column (3) includes both together, and column (4) gives the benchmark eigenvector regression. The downward curvature is concentrated in downstream centrality. In the downstream-only regression, the quadratic term is -1.092 ($p = 0.007$) with a turning point at 1.49. In the joint specification, the downstream quadratic remains large and negative at -0.998 ($p = 0.0003$), while the upstream quadratic is -0.618 with $p = 0.725$.

The downstream-upstream comparison is the key mechanism test. If the hump shape were just a generic feature of being important in the network, upstream and downstream centrality should look similar once both are entered together. Instead, the centrality margin that matters is exposure to downstream users, where the collective-action account locates the free-rider problem. A supplier with many customers faces larger policy stakes, but it also faces a larger set of industries that benefit from the lobbying campaign without needing to pay for it.

Table 2 Main mapped-data results: lobbying and network centrality

	(1) Downstream	(2) Upstream	(3) Joint	(4) Eigenvector
Panel A. Industry + Year FE (primary specification)				
Downstream centrality	3.262*** (1.218)		2.509*** (0.672)	
Downstream centrality ²	-1.092*** (0.391)		-0.998*** (0.264)	
Upstream centrality		4.361 (6.368)	4.136 (6.274)	
Upstream centrality ²		-0.680 (1.784)	-0.618 (1.752)	
Eigenvector centrality				5.745** (2.452)
Eigenvector centrality ²				-1.503** (0.653)
log (Value Added)	0.227 (0.389)	0.009 (0.374)	0.031 (0.376)	-0.038 (0.336)
Turning point	1.493 [1.018, 1.968]		1.257 [0.805, 1.709]	1.911 [1.629, 2.193]
Quadratic p (clustered)	0.007	0.704	0.000	0.024
Quadratic p (joint upstream)			0.725	
Wild-Webb hump p	0.243		0.281	0.334
Observations	1775	1775	1775	1775
Industries/years	71/25	71/25	71/25	71/25
Industry FE + Year FE	Yes	Yes	Yes	Yes

Dependent variable is $\log(1 + \text{mapped lobbying spend}_{it})$. Centrality measures are normalized within year to mean one. Standard errors clustered by industry are in parentheses. Turning-point 95% CI from industry-cluster bootstrap are in brackets. In column (3), the reported turning point, clustered quadratic p value, and Wild-Webb hump p value refer to downstream centrality. The separate row “Quadratic p (joint upstream)” reports the upstream quadratic term. Wild-Webb rows report one-sided hump p values from 9999 null-imposed industry-cluster draws with Webb six-point weights. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Figure 1 plots the same relationship nonparametrically. The curve rises through the middle of the distribution and declines for the most downstream-central industries. With only 71 industries, the upper tail is noisy, but the nonparametric fit is consistent with the quadratic specification.

The turning-point estimates map to recognizable industries. Chemical products (NAICS 325), credit intermediation (521CI), and insurance carriers (524) lie close to the peak of the downstream centrality curve. Miscellaneous professional services (5412OP) is the most downstream-central industry in the sample. On the fitted $\log(1 + \text{spend})$ scale, the lobbying prediction for that upper-tail observation is about 43% below the peak: the most downstream-central industries can create large policy spillovers, but they do not always spend the most.

4.2 Shape tests and coverage

Table 3 reports the stricter shape tests and the most direct coverage corrections for the downstream result. Panel A shows that the point estimates remain hump-shaped in both specifications: the downstream quadratic is negative and the turning point is interior. The formal pooled hump tests are more conservative. The implied peak-to-90th-percentile drop is modest, Webb hump p values are 0.243 and 0.281, and permutation p values are 0.300 and 0.240. The pooled estimates therefore remain nonmonotone, but they do not deliver a formally sharp hump result.

Panel B asks whether the same pattern survives simple corrections for incomplete coverage. The stabilized inverse-probability-weighted estimate keeps the downstream quadratic negative and slightly steeper than in the unweighted baseline, and the high-coverage

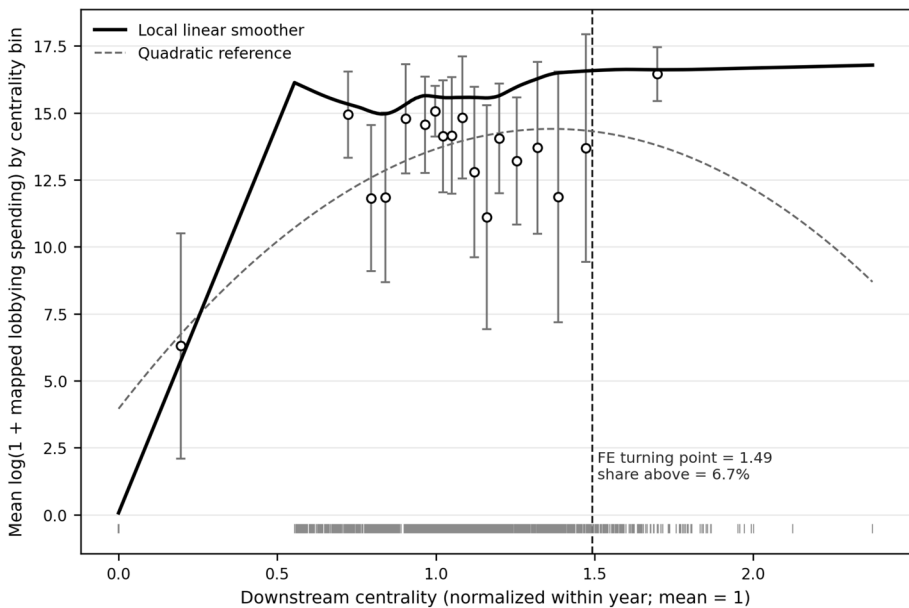


Fig. 1 Binned scatterplot of mapped lobbying against downstream centrality. Points are bin means of $\log(1 + \text{mapped spending})$ with 95% cluster-robust intervals. The solid line is a local linear smoother, and the dashed line is a quadratic reference

Table 3 Shape-focused inference for the downstream hump

Panel A. Direct tests of the hump shape						
Specification	Quadratic term	Cluster p	Wild-Webb hump p	Permutation hump p	Turning point [95% CI]	Peak-to-90th pct. drop [95% CI]
Downstream only	-1.092***	0.007	0.243	0.300	1.493 [0.906, 2.314]	0.006 [0.000, 0.432]
Joint downstream	-0.998***	0.000	0.281	0.240	1.257 [0.613, 2.186]	0.025 [0.000, 0.522]
Panel B. Observability correction and coverage						
Specification	Quadratic term	Cluster p	Turning point	Peak-to-90th pct. drop	Obs	
Unweighted baseline	-1.092	0.007	1.493	0.006	1775	
Stabilized IPW	-1.391	0.016	1.617	0.039	1437	
High coverage industries	-1.113	0.008	1.341	0.003	750	

Panel A treats the headline object as an interior hump rather than only a negative quadratic coefficient. The hump statistic equals the fitted drop from the turning point to the 90th percentile of downstream centrality when the estimated peak is interior, and zero otherwise. Wild-Webb p values use 9999 null-imposed industry-cluster draws. Permutation p values use 4999 industry-path permutations that reassign full downstream-centrality paths across industries while preserving each path's time structure. Panel B reports the unweighted baseline, the stabilized inverse-probability-weighting correction, and the split between industries above and below the median coverage rate. The peak-to-90th-percentile drop is the same hump statistic reported in Panel A. The weak common-support restriction is reported in Table S3 of the Electronic supplementary material. Tables S4 and S5 report graph-block sensitivity and year-by-year/meta evidence. Table S6 reports the broader control grid

industries show a similar pattern. A much tighter common-support restriction, reported in the Electronic supplementary material, changes the result because overlap is thin: once the sample falls to 144 covered observations, the quadratic turns positive. The Electronic supplementary material shows the same tradeoff from other angles. Negative curvature appears in every annual cross-section, but graph-bootstrap tail areas vary with the blocking rule, broader control sets weaken the pattern in some specifications, and the thin-overlap common-support sample is unstable. I therefore interpret the downstream hump as a stable mapped-spending pattern rather than as a formally sharp pooled shape result.

4.3 Mechanism evidence

Two pieces of evidence are consistent with the collective-action interpretation. First, Table 2 shows that the nonlinearity survives when downstream and upstream centrality are entered together, while the upstream quadratic is small and imprecise. Second, Table 4 compares the extensive and intensive margins using downstream centrality directly. The hump is visible on both margins, but it is economically much larger on the spending margin. Conditional on positive spending, the downstream quadratic is -1.207 with $p = 0.007$. On the extensive margin, the quadratic coefficient is only -0.052 , even though it is statistically detectable. This pattern is consistent with the view that downstream position matters mainly for how much already-organized industries spend once they are active, not for whether they appear in the data at all. The extensive-margin threshold connects to Bombardini (2008), who models firm-level lobby participation as a function of heterogeneous benefits from trade policy.

4.4 Eigenvector benchmark

The eigenvector specification plays a supporting role. It shows whether the hump also appears in a broad overall centrality measure and provides the benchmark regressor for Sect. 5, because eigenvector centrality is the most stable regressor under coverage loss. In column (4) of Table 2, the mapped-data estimate is concave, with $\hat{b}_2 = -1.503$ and clustered $p = 0.024$, but the corresponding Webb hump p value is 0.334. I therefore read the eigenvector result as supporting evidence rather than as a second standalone hump result.

Figure 2 shows the hump using the benchmark eigenvector measure. The fitted relationship peaks in the middle of the distribution, although less sharply than in the downstream

Table 4 Mechanism-consistent evidence: downstream entry versus spending

	Linear	Quadratic	p (Quad)	Observations
Extensive margin (linear probability model for positive spending)	0.144**	-0.052^{***}	0.003	1775
Intensive margin (log spending conditional on positive spending)	3.662***	-1.207^{***}	0.007	1437

Both rows use downstream centrality in industry and year fixed-effects models with industry-clustered standard errors. The linear and quadratic columns report the coefficients on downstream centrality and its square. The extensive-margin outcome is an indicator for positive mapped spending, while the intensive-margin row conditions on positive spending. The other mechanism-consistent fact appears in Table 2: once downstream and upstream centrality are entered together, the downstream quadratic remains large and negative while the upstream quadratic is small and imprecise. The extensive-margin coefficient is statistically detectable but economically much smaller than the intensive-margin coefficient. $***p < 0.01$, $**p < 0.05$, $*p < 0.1$

figure. The fitted magnitudes are sizeable: moving from the 25th to the 75th percentile of centrality raises the fitted index by 1.364 log points, which corresponds to a 3.91-times spending ratio. Moving from the 10th to the 90th percentile implies a 12.33-times ratio. At the estimated peak, 66 of the 71 industries by average centrality and about 93% of the industry-year observations lie on the increasing side of the curve. The hump therefore reflects an upper-tail decline, not a symmetric interior pattern.

This upper-tail reading translates the coefficients into political-economy language. The most central industries do not stop lobbying. Rather, they lobby less than the sectors just below them in the centrality ranking. In the model, that is the point at which broader downstream incidence stops being purely a source of private stakes and becomes a source of coordination loss. Economic centrality does not map one-for-one into political spending because the organizational technology weakens as the beneficiary coalition widens.

4.5 Stability

The stability exercises ask whether the downstream hump is driven by a few influential industries or a single year. Table 5 and Fig. 3 summarize how stable the downstream estimate is across straightforward perturbations. Dropping the top 5 or top 10 spenders barely changes the quadratic term. Leave-one-year-out estimation keeps the quadratic coefficient in the interval $[-1.236, -0.900]$ and the turning point in $[1.450, 1.546]$, so no single year drives the shape. As an additional temporal check, estimating the downstream specification separately over 1999–2006, 2007–2014, and 2015–2023 gives fitted turning points of 1.94, 1.81, and 1.91, and an interaction test does not reject stability of the downstream profile across subperiods ($p = 0.463$). Combined with the year-by-year sign summary in

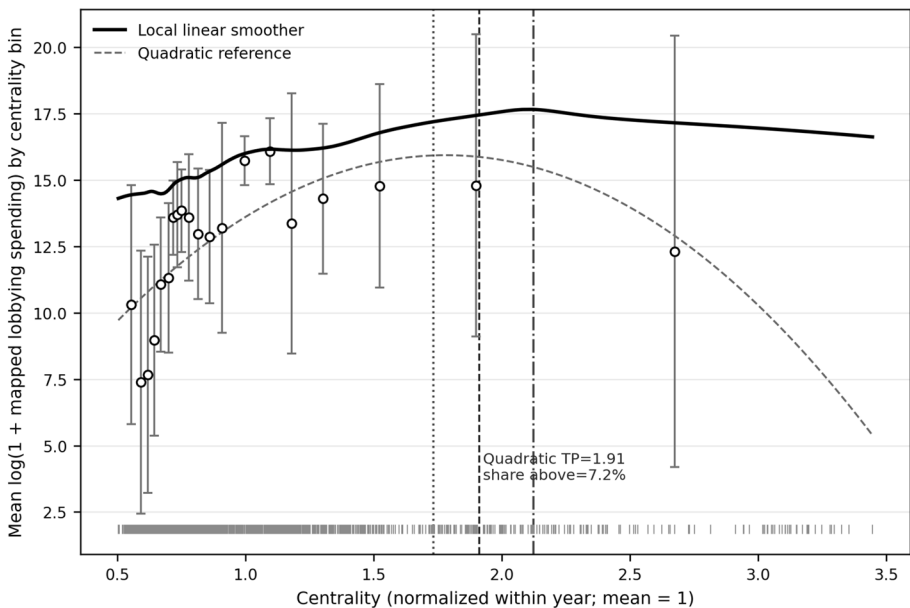


Fig. 2 Binned scatterplot of mapped lobbying against overall eigenvector centrality. The figure summarizes the same hump-shaped pattern shown parametrically in the fixed-effects regressions

Table 5 Stability of the downstream specification

Specification	Quadratic term	<i>p</i> value	Turning point
Baseline FE	-1.092***	0.007	1.493
Drop top 5 spenders	-1.087***	0.008	1.512
Drop top 10 spenders	-1.026**	0.012	1.586
Leave-one-year-out	[-0.900, -1.236]	—	[1.450, 1.546]

All rows use the downstream-centrality fixed-effects specification with industry and year fixed effects and log(Value Added) as a control. The leave-one-year-out row reports the range across the 25 yearly re-estimates. The corresponding path is shown in Fig. 3. Table S5 in the Electronic supplementary material summarizes the complementary year-by-year cross-sectional evidence. *** $p < 0.01$, ** $p < 0.05$

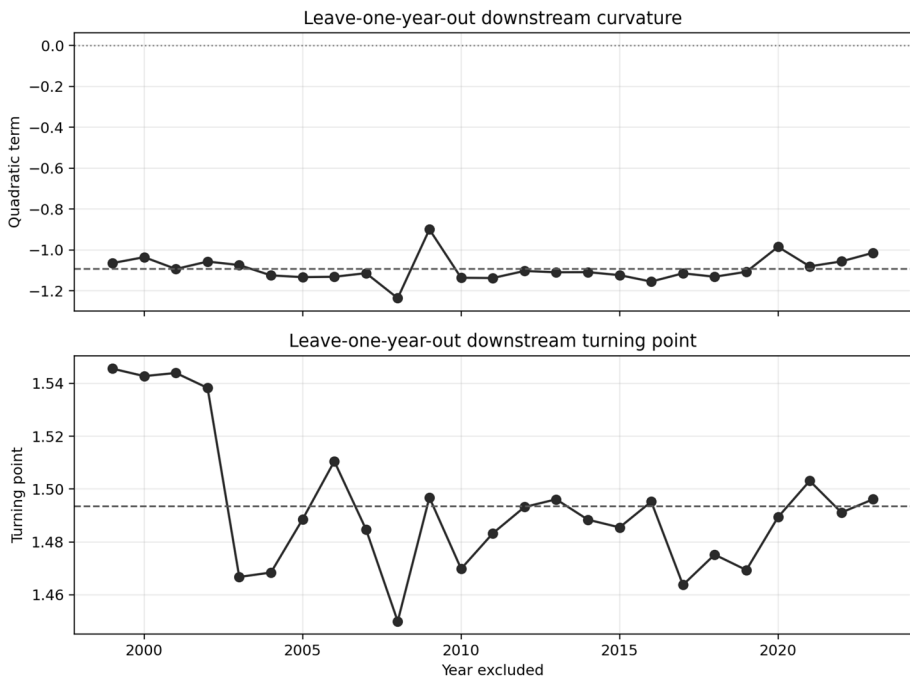


Fig. 3 Leave-one-year-out downstream stability. The top panel reports the downstream quadratic coefficient after excluding each year in turn. The bottom panel reports the corresponding turning point. Dashed lines mark the full-sample estimate

Table S5 of the Electronic supplementary material, the stability evidence suggests that the downstream hump is persistent as a mapped-data sign pattern even if the pooled shape tests remain conservative.

5 Unmapped spending

The mapped-data results document the main observed pattern: mapped spending is hump-shaped in downstream centrality. This section asks how much the benchmark eigenvector result can change once the unmapped 68% of spending is taken into account. The exercise

does not resolve attribution for the downstream result, and it does not establish that total lobbying is hump-shaped.

Because the regressor is observed even when the outcome is not, this is not a generic missing-outcome problem. For each year, the data provide mapped spending by industry, total unmapped spending, and the network position of every industry. The unobserved object is the allocation of the unmapped block across industries. The Electronic supplementary material formalizes that problem and shows how the observed network restricts the exercise. I do not report the unrestricted Manski (2003) envelope because it is too wide to be useful relative to the structured allocations reported below. The exercise is a structured partial-identification analysis rather than an unrestricted missing-data envelope.

I start from three anchor allocations for the unmapped mass: uniform, proportional to mapped spending, and proportional to industry size. These are transparent benchmarks anchored in the observed spending distribution and an economically natural size measure. Table 6 reports that all three preserve negative curvature in the benchmark full-network specification. The same table reports the preferred κ -band bounds around the size-proportional allocation. Under the narrow B1 band, the identified set for the curvature parameter on the standardized full-network scale is $[-0.5885, +0.5098]$. Under the wider B2 band, it becomes $[-1.0742, +0.9520]$. Table S7 in the Electronic supplementary material reports the empirical calibration behind those band widths. Negative curvature survives natural allocation rules, but the preferred structured intervals include zero.

The interval is asymmetric in practice. Inside the B1 feasible set, 99.4% of sampled allocations still deliver a negative quadratic term, and even under B2 the corresponding share is 91.7%. Figure 4 reports the same comparison visually. The sign-reversing corner shifts spending toward low-centrality industries and away from the most central ones, which

Table 6 Bounds summary for unmapped lobbying spending (benchmark eigenvector regressor)

Panel A. Anchor allocations and κ -band bounds (X1 full-network centrality)			
Assumption	$\tilde{\beta}_2$ or interval	Share with $\tilde{\beta}_2 < 0$	Turning point or band
Mapped-only reference	-0.411	1.000	1.72
Uniform allocation	-0.022	1.000	1.66
Proportional to mapped spending	-0.431	1.000	1.75
Size-proportional allocation	-0.036	1.000	2.28
Preferred B1 $\kappa \in [0.5, 1.5]$	$[-0.5885, +0.5098]$	0.994	$[1.57, 4.96]$
B2 $\kappa \in [0.25, 2.0]$	$[-1.0742, +0.9520]$	0.917	$[1.12, 7.63]$
Panel B. Characterizing the B1 sign-reversing corner			
Centrality quintile	Mean centrality	$a^{\max}/a^{\text{ref}}$	
Q1 (lowest)	-0.7687	1.0679	
Q2	-0.5724	1.0945	
Q3	-0.3852	0.8847	
Q4	0.0851	1.0915	
Q5 (highest)	1.6411	0.9015	

Panel A reports the curvature parameter on the year-standardized full-network centrality scale used in the bounds pipeline. The mapped-only row is the reference estimate on that scale. For deterministic anchor allocations, the share with $\tilde{\beta}_2 < 0$ is either zero or one by construction. For the B1 and B2 rows, it is the share of sampled admissible allocations with negative curvature. Panel B reports the allocation ratios for the B1 upper-bound solution relative to the size-proportional benchmark. A ratio above one means that the sign-reversing corner assigns more spending to that quintile than the size-proportional benchmark, while a ratio below one means less. Table S7 in the electronic material reports the analogous directional downstream exercise and the κ -band calibration details

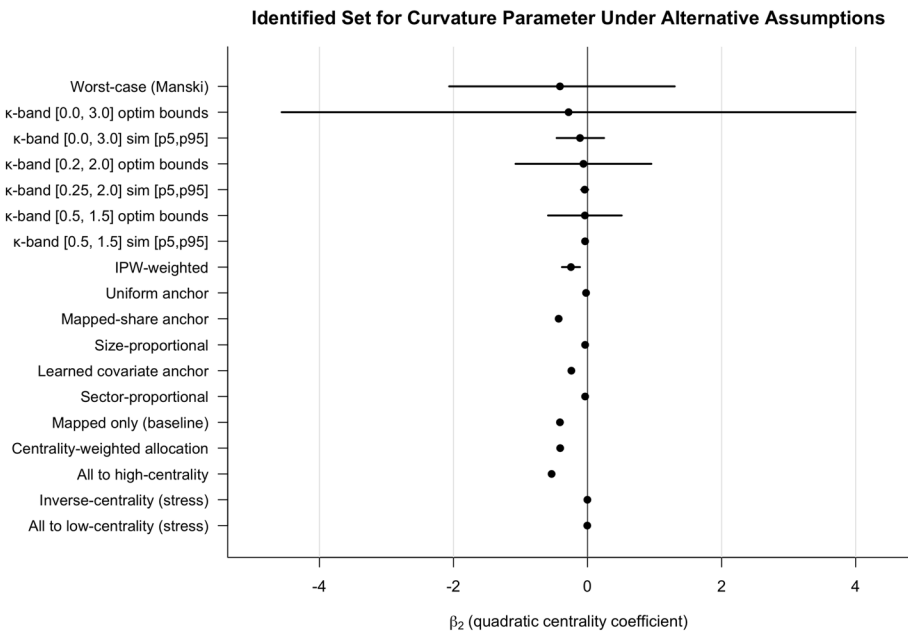


Fig. 4 Identified set for the curvature coefficient under alternative allocation assumptions. Black dots denote point estimates and horizontal black segments denote intervals under the alternative allocation assumptions. The vertical reference line marks zero

conflicts with the mapped spending data. This asymmetry gives economic content to the bounds: reversing the sign requires allocating the unmapped mass in a way that pushes against the observed pattern. The structured interval nevertheless includes zero.

The bounded exercise therefore sharpens, but does not overturn, the mapped-data result. The downstream hump remains the main empirical pattern in observed spending. The same attribution exercise is less informative for directional downstream centrality because that regressor becomes less stable when the covered network is recomputed. For the benchmark overall centrality measure, economically motivated allocations continue to produce negative curvature, but the preferred admissible band still includes zero. The structured allocations make a sign reversal difficult, but they do not rule it out.

6 Conclusion

The paper's main result is a clear empirical pattern in mapped industry lobbying. Lobbying is hump-shaped in downstream centrality: industries with moderate downstream reach spend more than both peripheral industries and the most downstream-central suppliers. That pattern appears in the pooled fixed-effects estimates, the annual cross-sections, the intensive-margin results, and a range of stability exercises.

This pattern matters because downstream position changes both the return to lobbying and the scope for free-riding. Industries with broader downstream reach have more at stake, but they also create a larger set of beneficiaries who can gain from lobbying without pay-

ing for it. The evidence points in that direction because the nonlinearity is concentrated in downstream rather than upstream centrality, and because it is much larger on the spending margin than on the entry margin.

The main data constraint is on the outcome, not on the network side of the design. Only part of total reported lobbying can be mapped cleanly to industries, so the strongest claim is about observed mapped spending. For this reason, the paper pairs the downstream results with observability corrections and structured bounds for the benchmark eigenvector measure. Those exercises qualify the strength of the attribution claim, but they do not overturn the central mapped-data pattern.

The mechanism is most likely to apply in settings with transparent lobbying records, organized industries, and production networks in which policy changes propagate through supplier-customer links. Advanced economies with formal lobbying disclosure and active industry associations are the closest analogues. The data exercise is more U.S.-specific because the LDA reporting system, client identities, and treatment of associations determine which expenditures can be mapped to industries.

The measurement problem also has a transparency implication. If highly downstream industries increasingly act through trade associations, umbrella groups, or other shared vehicles, client-level lobbying records may obscure the economic incidence of political activity. The issue is not only whether lobbying is reported, but whether reported expenditures can be attributed to the industries whose members benefit from the policy. More information on association membership and industry composition would make it easier to distinguish lower lobbying from lobbying that is harder to attribute.

Several within-U.S. extensions would help evaluate the mechanism more directly. Firm-level production-network data linked to LobbyView would show whether the industry-level pattern reflects aggregation of heterogeneous firm incentives. Event studies around sector-specific policy changes would connect annual spending patterns to identifiable policy opportunities. A further step is to link issue-specific lobbying to policy outcomes, which would ask whether downstream position predicts political success as well as spending.

The broader implication is that the industries at the center of supply chains are not automatically the industries that lobby the most. Political organization depends not only on economic importance, but also on whether an industry can privately appropriate the gains from policy influence.

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Data availability This study uses existing data rather than newly generated data. The analysis combines harmonized U.S. lobbying data from LobbyView with U.S. input–output data. A replication package accompanies the paper and contains the data and code needed to reproduce the results.

Declarations

Conflict of interest The authors declare no conflict of interest.

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